Akshat Katiyar

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Senior Backend Engineer — AI & Platform

Summary

Senior Backend Engineer with 7+ years shipping production Python and Go services, REST APIs, and PostgreSQL-backed platforms end to end. Builds GenAI products (RAG, LLM agents, OpenAI, LiteLLM/Portkey routing, Qdrant) and uses Claude daily to accelerate delivery while maintaining code-review discipline. Proven in global, regulated environments (banking, observability) with strong DevOps and full-stack (Angular) delivery.

Core Competencies

- Backend (Python, FastAPI, Flask, Go) — PostgreSQL, SQLAlchemy — REST, gRPC, Microservices
- GenAI & LLMs (LangChain, OpenAI, LiteLLM, Portkey, RAG, Qdrant) — Claude/Cursor AI-assisted development — MCP
- DevOps (Docker, K8s, CI/CD, Ansible, Terraform) — AWS — Angular — Full-stack delivery

Technical Skills

- **Languages:** Python (FastAPI, Flask, Django, asyncio), Go, JavaScript, TypeScript, SQL
- **Backend & Data:** REST/gRPC APIs, PostgreSQL, SQLAlchemy, MongoDB, Redis, DynamoDB, RabbitMQ, System Design
- **AI/ML:** Claude, Cursor, OpenAI, LangChain, LiteLLM, Portkey, RAG, LLMs, Prompt Engineering, FAISS, Qdrant, MCP, TensorFlow, PyTorch
- **DevOps & Cloud:** Docker, Kubernetes, CI/CD (Jenkins, TeamCity, GitHub Actions), Ansible, Terraform, Airflow, AWS (Lambda, S3, RDS, SQS, EKS, SageMaker, Bedrock)
- **Observability & Tools:** OpenTelemetry, F5, LDAP, ZeroMQ, Agile/Scrum

Work Experience

Senior Software Engineer **CBA India (Commonwealth Bank), Bengaluru** **May 2023 – Present**

- Prototyped a GenAI network chatbot with LiteLLM multi-model routing and Portkey observability, cutting domain Q&A resolution time by 30% for network operations.
- Built and published 'cbapysnow', a reusable Python package for ServiceNow integration that reduced script development time by 25% across teams.
- Led backend architecture and API design in Python and Go for business-critical platforms; shipped features through code review, testing, and release cycles.
- Delivered F5 automation POC (FastAPI, Airflow, AWS) end to end, demonstrating a 40% reduction in manual operational tasks.
- Shipped F5Portal (Python backend, JavaScript UI) end to end, reducing operational data retrieval time by 30%.

Software Engineer III **AppDynamics (Cisco), Bengaluru** **Nov 2022 – May 2023**

- Owned production-quality improvements on the Python Agent, shipping fixes for customer-facing observability at scale.
- Built Ansible-based agent lifecycle automation (install/upgrade/rollback/uninstall) across VMs, cutting deployment time by 50%+ and improving release reliability.
- Built high-throughput IPC with ZeroMQ between core services, improving data throughput by 20% for real-time monitoring pipelines.

Software Engineering Specialist **Societe Generale Solution Center, Bengaluru** **Jan 2019 – Nov 2022**

- Promoted to Senior Software Engineer (July 2022) for consistent delivery and technical leadership.
- Built and deployed FTS Portal (Angular frontend, Flask REST APIs, PostgreSQL) end to end, used by teams across 6 global regions for ticket escalation and async handoffs.
- Architected Alert Management, an ML-driven application for proactive incident detection, reducing MTTR for critical incidents by 25%.
- Developed Marley Automation (Ansible-backed APIs, Jenkins CI/CD, Kibana dashboards), automating 80%+ of routine Cisco switch configuration tasks.

Selected Projects

GenAI Document Q&A (RAG) **Python, LangChain, OpenAI, LiteLLM, Portkey, FAISS, Qdrant**
Built a RAG pipeline with LangChain, OpenAI embeddings, FAISS and Qdrant retrieval, and LiteLLM/Portkey for model routing and gateway observability. Evaluated answer relevance end to end. *85% better relevance on complex queries; 60% faster information retrieval.*

Scalable Event Processing Service **Go, AWS Lambda, SQS, DynamoDB**
Serverless backend ingesting events via SQS, transforming in Lambda, persisting to DynamoDB. *100k events/day, under 200ms latency, 99.99% uptime.*

Cloud Log Analysis Pipeline **Go, AWS S3, Elasticsearch, Kibana**
Scalable log ingestion and search backend with ELK visualization. *40% faster log queries vs. prior system.*

Education

Master of Technology, Software Engineering **Birla Institute of Technology And Science - Pilani, Rajasthan**
Jul 2021 – Jun 2023

Bachelor of Engineering, Computer Science **PSG College of Technology - Coimbatore, Tamil Nadu**
Aug 2014 – May 2019